

ANGSHUMAN MAZUMDAR

School of Applied and Creative Computing

Purdue University, West Lafayette, IN

amazum619@gmail.com

<https://binaryvectorjr.github.io/>

PROFESSIONAL SUMMARY

PhD candidate and researcher specializing in agent-based modelling, real-time simulation, physics-based modeling, and high-performance data driven computing. Experienced in C++, Unity, Unreal Engine, Raylib, GPU programming, and multi-agent systems. Research primarily focuses on simulation algorithms, game engine tools, and interactive system design, and curriculum development for these topics.

RESEARCH INTERESTS

Agent Based Modelling Simulation, Simulation for Education, Game Design and Development Education, Change Management Based Curriculum Design

TECHNICAL SKILLS

- **Programming Languages:** C++, C#, CUDA, OpenGL, Vulkan, VBA, Python, HTML
- **Engines & Frameworks:** Unity (Desktop & VR), Unreal Engine (Blueprint & C++), Raylib
- **Simulation:** NetLogo, AnyLogic, Houdini
- **Modelling:** Siemens NX, Solidworks, Blender
- **Data & Workflow Tools:** Teamcenter PLM, MS Power Automate, Git, Jira, Qualtrics, NXOpen, Solidworks API
- **Documentation & Technical Communication:** Doxygen, Overleaf, Markdown, Confluence
- **Soft Skills:** Mentorship, Leadership, Teaching
- **Research Skills:** Participatory Design, Heuristic Evaluation, Quantitative Analysis, User Research, Descriptive Statistical Analysis, Social and Behavioral Research Experimental Design, Mixed Methods Design

EDUCATION

Ph.D. in Technology (2021 - 2026)

Purdue University, West Lafayette, IN

M.S. in Computer Graphics Technology (2019 - 2021)

Purdue University, West Lafayette, IN

Semester Abroad Program (Capstone in Robotics) (2019)

Rochester Institute of Technology, Rochester, NY

B. Tech. in Electronics and Communication Engineering (2015 - 2019)

Vellore Institute of Technology, Vellore, Tamil Nadu

RESEARCH EXPERIENCE

Games Innovation Laboratory – Purdue University (08/2021 – Present)

Graduate Mentor and Research Advisor

- Guided multidisciplinary teams in VR development, simulation design, and real-time graphics research.
- Advised projects involving agent AI, optimization algorithms, and immersive experience design.

INDUSTRY EXPERIENCE

Bharti Airtel (05/2018 - 06/2018)

Summer Internship (Networking Department)

- Developed automated workflows for data collection and validation.
- Built VBA tools to reduce manual processing and improve reliability.

TEACHING EXPERIENCE

School of Engineering Technology – Purdue University (08/2020 – Present)

Lead Graduate Teaching Assistant (NX, Aras, & Teamcenter)

- Led a team of 10 TAs supporting 1000+ students across engineering graphics and PLM courses.
- Built Unity, VBA, and Power Automate-based tools to reduce grading time by 50%.
- Designed frameworks for continuous improvement (CM2) in engineering graphics education.
- Recognized with the **Graduate Teaching Award** by the Purdue University Teaching Academy

PROJECTS

- [Vasoo Virtual Stadium Experience](#)
Iconic Engine (2022)
Interaction Mechanic Developer & Project Manager
Implemented core interaction systems for large-scale virtual stadium environments.
- [Sonata Theory \(Steam\)](#)
MAYJ Studios (2021)
Audio Producer and Asset Pipeline Engineer
Contributed to audio systems and production pipeline for a shipped indie game.

PUBLICATIONS

Full Papers (Peer Reviewed)

- [Enterprise PDM as Digital Backbone in a Large First-Year Engineering Course](#)
Fuerst, T., Camba, Jorge., Mazumdar, A., Koellisch, J. ASEE 2025 (Quebec, Canada)
Formalized PLM/PDM and CM2-based curricula for engineering education and digital enterprise integration.
- [Engineering Graphics Education for the Digital Enterprise](#)
Fuerst, T., Camba, J., Mazumdar, A. ASEE 2024 (Portland, OR)
Introduced project-based learning and continuous improvement frameworks in engineering graphics courses.
- [Synthesizing Affective Virtual Reality Multicharacter Experiences \(Master's Thesis\)](#)
Mazumdar, A., Mousas, C. Journal of Computer Animation & Virtual Worlds (2021)
Developed optimization algorithms using Markov Chain Monte Carlo and Simulated Annealing for affective VR experiences.
- [Virtual reality game level layout design for real environment constraints](#)
Liu, H., Wang, Z., Mazumdar, A., Mousas, C. Journal of Graphics and Visual Computing (2021)
Built ECS based intelligent patrol agents with zone-based detection, visibility-cost modeling, and follow intelligence for Oculus Quest.
- [Walking in a Crowd Full of Virtual Characters: Effects of Virtual Character Appearance on Human Movement Behavior](#)
Nelson, M. G., Mazumdar, A., Jamal, S., Chen, Y., Mousas, C. ISVC 2020 (San Diego,

CA)

Created pipelines for optimizing Mixamo assets and integrating them into Unity for crowd simulation studies.

- [Parkinson's Disease Assist Device Using Machine Learning and Internet of Things](#)
Baby, C. J., Mazumdar, A., Sood, H., Gupta, Y., Panda, A., Poonkuzhali, R. ICCSP 2018 (Chennai, India)
Created CAD data and stress simulation data for a prototype of a spoon which senses the hand motion of an individual suffering from Parkinson's disease.

Workshop Papers (Peer Reviewed)

- [Embodiment for the Difference: A VR Experience of Bipolar Disorder](#)
Dong, J., Mazumdar, A., Yang, F. C., Chen, H., Han, S., Mousas, C. IEEEVR 2020 (Atlanta, GA)
Developed an immersive simulation embodying bipolar disorder for the 3DUI Contest.

HONORS & AWARDS

- Purdue Graduate Teaching Award (2025)
- Purdue Polytechnic Graduate School Summer Research Grant (2023)
- Daniel & Martina Lewis Graduate Endowment Scholarship (2020) (\$3500)

PRESENTATIONS & INVITED TALKS

- Building a Digital Thread Discipline at Purdue (ACE 2023, Orlando, FL)
- Computer Animation & Social Agents Conference (CASA 21, Ottawa, Canada)
- TECH 662: Philosophy of Technology Guest Lecture (Agent Based Modelling)

DIVERSITY, EQUITY, & INCLUSION EFFORTS

- Graduate Mentor for Undergraduate Students, Games and Innovation Laboratory (2021-2025)
- Technology Advisor, DESS PDM Club (Purdue) (2024-2025)
- Graduate Mentor for Undergraduate Game Development Teams, Game Development Practicum Course (2021)
- Technology Advisor, Game Developers United (Purdue) (2021)
- Member and Dungeon Master, Dungeons & Dragons Weekend Cohort (2020-2021)
- Events and Promotional Media Advisor, Movement of Expression (MOX), (2018-2019)
- Graphics Design Lead, Vimaanans (VIT RC Aero Club) (2017-2019)

LANGUAGES

Assamese (Native), English (Full-Working), Hindi (Full-Working), Spanish (Novice)

FEATURED NEWS ARTICLES

- [Polytechnic grad students design the future of Purdue sports in the metaverse](#)
- [New student-run digital enterprise group manages engineering projects for clubs across Purdue](#)